Temporal Data Fusion: Working Problem Setup

1 Motivation

In many settings, researchers observe multiple data sources with different strengths. One data source may be high-frequency and individual-level, such as an online survey, but biased because respondents are not representative of the target population. Another data source may be closer to the target population, such as administrative records or benchmark survey data, but may only be available at an aggregate level or with a time delay. The goal is to combine these sources to estimate current target-population quantities. The main challenge is that the most recent data may come from the biased source, while the more reliable target-population data may only be available for earlier time periods.

2 Problem Setting

For each time $t = 1, \ldots, T$, let S_t denote a biased high-resolution source distribution at time t. In many contexts, S_t may represent individual-level survey data for a health indicator of interest, such as vaccination status or Medicaid enrollment. Let P_t denote the corresponding low-resolution, low-bias target population distribution at time t.

We view S_t as the respondent distribution induced from P_t through a response mechanism. Let $R_t \in \{0, 1\}$ denote a response indicator at time t, where $R_t = 1$ indicates if a unit responded to the survey and is included in the biased source. The probability of response can be modeled as

$$\alpha_t(x, y) = \mathbb{P}_{P_t}(R_t = 1 \mid X = x, Y = y).$$

Thus, S_t is the distribution of (X, Y) among respondents at time t.

Because the probability of response may depend on covariates X, the outcome Y, or unobserved factors related to Y, the respondent distribution S_t may differ from the target population distribution P_t. The data availability is temporal. At the current time T, we observe individual-level samples from the biased respondent distribution through time T:

$$(X_{it}^S, Y_{it}^S) \sim S_t, \qquad i = 1, \ldots, n_t^S, \qquad t = 1, \ldots, T.$$

The biased source S_t is observed up to the current time T:

$$S_1, \ldots, S_T.$$

In contrast, the target population source P_t is only observed up to a lagged time $T - k$:

$$P_1, \ldots, P_{T-k}.$$

For $t = 1, \ldots, T - k$, we observe individual-level target population samples. The goal is to estimate a current target-population quantity, such as the mean outcome

$$\mu_T = \mathbb{E}_{P_T}[Y],$$

or a subgroup-level quantity

$$\mu_{g,T} = \mathbb{E}_{P_T}[Y \mid X \in g],$$

by constructing a candidate target-population distribution Q_T at the current time that preserves useful high-resolution information from the recent biased source S_T while remaining consistent with low-resolution target population information.

3 Covariate Free Setup: Binary Outcome

In a simplified setting, we assume no covariates and a binary outcome,

$$Y_i \in \{0, 1\}.$$

At each time t,

$$(Y_i, R_i) \sim P_t,$$

where $R_i \in \{0, 1\}$ is a response indicator. The biased survey distribution is

$$S_t = \mathcal{L}_{P_t}(Y_i \mid R_i = 1).$$

For binary outcomes, a simple response-bias parameter is the ratio

$$\alpha_t = \frac{\mathbb{P}_{P_t}(R_i = 1 \mid Y_i = 1)}{\mathbb{P}_{P_t}(R_i = 1 \mid Y_i = 0)}.$$

We assume that

$$\alpha_t = \alpha$$

for all t. Let

$$p_t = \mathbb{P}_{P_t}(Y_i = 1)$$

and

$$s_t = \mathbb{P}_{P_t}(Y_i = 1 \mid R_i = 1).$$

We solve for $\mathbb{P}_{P_t}(Y_i = 1)$. By Bayes' rule,

$$s_t = \frac{\mathbb{P}_{P_t}(R_i = 1 \mid Y_i = 1)\mathbb{P}_{P_t}(Y_i = 1)}{\mathbb{P}_{P_t}(R_i = 1)}.$$

Using the law of total probability,

$$s_t = \frac{\mathbb{P}_{P_t}(R_i = 1 \mid Y_i = 1)\mathbb{P}_{P_t}(Y_i = 1)}{\mathbb{P}_{P_t}(R_i = 1 \mid Y_i = 1)\mathbb{P}_{P_t}(Y_i = 1) + \mathbb{P}_{P_t}(R_i = 1 \mid Y_i = 0)\,(1 - \mathbb{P}_{P_t}(Y_i = 1))}.$$

From the definition of the response-bias parameter,

$$\mathbb{P}_{P_t}(R_i = 1 \mid Y_i = 1) = \alpha_t \mathbb{P}_{P_t}(R_i = 1 \mid Y_i = 0).$$

Substituting this into the expression above gives

$$s_t = \frac{\alpha_t \mathbb{P}_{P_t}(R_i = 1 \mid Y_i = 0)\mathbb{P}_{P_t}(Y_i = 1)}{\alpha_t \mathbb{P}_{P_t}(R_i = 1 \mid Y_i = 0)\mathbb{P}_{P_t}(Y_i = 1) + \mathbb{P}_{P_t}(R_i = 1 \mid Y_i = 0)\,(1 - \mathbb{P}_{P_t}(Y_i = 1))}.$$

Canceling out $\mathbb{P}_{P_t}(R_i = 1 \mid Y_i = 0)$, we obtain

$$s_t = \frac{\alpha_t \mathbb{P}_{P_t}(Y_i = 1)}{\alpha_t \mathbb{P}_{P_t}(Y_i = 1) + (1 - \mathbb{P}_{P_t}(Y_i = 1))}.$$

This becomes

$$s_t = \frac{\alpha_t p_t}{\alpha_t p_t + (1 - p_t)}.$$

Under the assumption that $\alpha_t = \alpha$, we have

$$s_t = \frac{\alpha p_t}{\alpha p_t + (1 - p_t)}.$$

Solving for p_t, we get

$$p_t = \frac{s_t}{\alpha + s_t - \alpha s_t}.$$

At time t, if we know α and observe s_t, then we can recover p_t. For the current time T, we can solve

$$p_T = \frac{s_T}{\alpha + s_T - \alpha s_T}.$$

In past periods where both S_t and P_t are observed, we can estimate both s_t and p_t. Rearranging

$$s_t = \frac{\alpha p_t}{\alpha p_t + (1 - p_t)}$$

gives

$$\alpha = \frac{s_t(1 - p_t)}{p_t(1 - s_t)}.$$

Therefore, past paired observations from S_t and P_t can be used to estimate α, which can then be used to recover the current target mean p_T from the current survey mean s_T.

$$\begin{aligned}
p_t - s_t &= \frac{s_t}{\alpha + s_t - \alpha s_t} - s_t \\
p_t - s_t &= \frac{s_t - s_t(\alpha + s_t - \alpha s_t)}{\alpha + s_t - \alpha s_t} \\
p_t - s_t &= \frac{s_t - s_t(\alpha + s_t(1 - \alpha))}{\alpha + s_t - \alpha s_t} \\
p_t - s_t &= \frac{s_t(1 - \alpha) - s_t^2(1 - \alpha)}{\alpha + s_t - \alpha s_t} \\
p_t - s_t &= \frac{s_t(1 - s_t)(1 - \alpha)}{\alpha + s_t(1 - \alpha)} \\
p_t &= s_t + \frac{s_t(1 - s_t)(1 - \alpha)}{\alpha + s_t(1 - \alpha)}
\end{aligned}$$

Since

$$p_t - s_t = \frac{s_t(1 - s_t)(1 - \alpha_t)}{\alpha_t + s_t(1 - \alpha_t)},$$

and

$$\alpha_t + s_t(1 - \alpha_t) = \alpha_t(1 - s_t) + s_t > 0$$

for every $\alpha_t > 0$ and $0 \le s_t \le 1$, the sign of $p_t - s_t$ is determined entirely by

$$s_t(1-s_t)(1-\alpha_t).$$

For $0 < s_t < 1$, we have $s_t(1-s_t) > 0$. Therefore,

$$\begin{aligned}
\alpha_t = 1 &\implies p_t - s_t = 0 \implies p_t = s_t, \\
\alpha_t > 1 &\implies 1-\alpha_t < 0 \implies p_t - s_t < 0 \implies p_t < s_t, \\
0 < \alpha_t < 1 &\implies 1-\alpha_t > 0 \implies p_t - s_t > 0 \implies p_t > s_t.
\end{aligned}$$

Recall that

$$\alpha_t = \frac{\mathbb{P}_{P_t}(R_i = 1 \mid Y_i = 1)}{\mathbb{P}_{P_t}(R_i = 1 \mid Y_i = 0)}.$$

Thus, for $0 < s_t < 1$,

$$p_t \begin{cases} = s_t, & \alpha_t = 1, \\ < s_t, & \alpha_t > 1, \\ > s_t, & 0 < \alpha_t < 1. \end{cases}$$

At the boundary values,

$$p_t = \begin{cases} 0, & s_t = 0, \\ 1, & s_t = 1, \end{cases} \qquad \text{for every } \alpha_t > 0.$$

4 Covariate Free Setup: Continuous Outcome

In this setting we assume no covariates, but now the outcome $y \in \mathbb{R}$. $f_{P_T}(y)$ is the density of Y in the target population source and $f_{S_T}(y) = f_{P_T}(y|R=1)$ is the density of Y in the biased survey source data.

$$f_{S_T}(y) = f_{P_T}(y|R=1)$$

Using bayes' rule:

$$f_{S_T}(y) = f_{P_T}(y|R=1) = \frac{\mathbb{P}_{P_T}(R=1|Y=y) f_{P_T}(y)}{\mathbb{P}_{P_T}(R=1)}$$

The probability of response for the continuous, covariate free case is $\alpha_T(y) = \mathbb{P}_{P_T}(R=1|Y=y)$. We can make the following substitution:

$$f_{S_T}(y) = \frac{\mathbb{P}_{P_T}(R=1|Y=y) f_{P_T}(y)}{\mathbb{P}_{P_T}(R=1)} = \frac{\alpha_T(y) f_{P_T}(y)}{\mathbb{P}_{P_T}(R=1)}$$

$f_{S_T}(y) = \frac{\alpha_T(y) f_{P_T}(y)}{\mathbb{P}_{P_T}(R=1)}$ can be rewritten as:

$$f_{P_T}(y) = \frac{f_{S_T}(y)\mathbb{P}_{P_T}(R=1)}{\alpha_T(y)}$$

$\mathbb{P}_{P_T}(R=1)$ is unknown, but can be rewritten as so:

$$\int f_{P_T}(y)dy = 1$$

$$\int \frac{f_{S_T}(y)\mathbb{P}_{P_T}(R=1)}{\alpha_T(y)}dy = 1$$

$$\mathbb{P}_{P_T}(R=1)\int \frac{f_{S_T}(y)}{\alpha_T(y)}dy = 1$$

$$\mathbb{P}_{S_T}(R=1) = \frac{1}{\int \frac{f_{S_T}(y)}{\alpha_T(y)}dy}$$

substituting this back into $f_{P_T}(y)$:

$$f_{P_T}(y) = \frac{f_{S_T}(y)\mathbb{P}_{P_T}(R=1)}{\alpha_T(y)} = \frac{f_{S_T}(y)}{\alpha_T(y)\int \frac{f_{S_T}(y)}{\alpha_T(y)}dy}$$

The target quantity is:

$$p_t = E_{P_T}[Y] = \int y f_{P_T}(y)dy = \int y \frac{f_{S_T}(y)}{\alpha_T(y)\int \frac{f_{S_T}(y)}{\alpha_T(y)}dy}dy$$

$$= \frac{1}{\int \frac{f_{S_T}(y)}{\alpha_T(y)}dy}\int y\frac{f_{S_T}(y)}{\alpha_T(y)}$$

Because of $E_{S_T}[g(Y)] = \int g(y)f_{S_T}(y)dy$:

$$p_t = \frac{1}{\int \frac{f_{S_T}(y)}{\alpha_T(y)}dy}\int y\frac{f_{S_T}(y)}{\alpha_T(y)} = \frac{E_{S_T}[\frac{y}{\alpha_T(y)}]}{E_{S_T}[\frac{1}{\alpha_T(y)}]}$$

We can show that the target population estimator can also be written as a loss-minimization problem:

$$\min_{\theta\in\mathbb{R}} E_{S_T}[L(Y;\theta)],$$

where

$$L(Y;\theta) = \frac{1}{\alpha_T(Y)}\left(\theta^2 - 2\theta Y\right).$$

By the stationary sampling bias assumption,

$$\frac{\alpha_T(y)}{\alpha_T(y')} = \frac{\alpha_{T-1}(y)}{\alpha_{T-1}(y')}.$$

Rearranging,

$$\frac{1}{\alpha_T(y)} = \frac{\alpha_{T-1}(y')}{\alpha_{T-1}(y)\alpha_T(y')}.$$

By Bayes' rule,

$$\alpha_{T-1}(y) = \frac{f_{S_{T-1}}(y)}{f_{P_{T-1}}(y)}\mathbb{P}_{P_{T-1}}(R=1).$$

Therefore,

$$\frac{1}{\alpha_T(y)} = \frac{\alpha_{T-1}(y')}{\alpha_T(y')\mathbb{P}_{P_{T-1}}(R=1)} \cdot \frac{f_{P_{T-1}}(y)}{f_{S_{T-1}}(y)}.$$

Let

$$c = \frac{\alpha_{T-1}(y')}{\alpha_T(y')\mathbb{P}_{P_{T-1}}(R=1)}.$$

Then

$$\frac{1}{\alpha_T(y)} = c\frac{f_{P_{T-1}}(y)}{f_{S_{T-1}}(y)}.$$

Substituting into the loss gives

$$E_{S_T}\left[\frac{1}{\alpha_T(Y)}\left(\theta^2 - 2\theta Y\right)\right] = cE_{S_T}\left[\frac{f_{P_{T-1}}(Y)}{f_{S_{T-1}}(Y)}\left(\theta^2 - 2\theta Y\right)\right].$$

Since c does not depend on θ, it does not affect the minimizer. Therefore, we can solve the following using sample estimators:

$$\min_{\theta\in\mathbb{R}} E_{S_T}\left[\frac{f_{P_{T-1}}(Y)}{f_{S_{T-1}}(Y)}\left(\theta^2 - 2\theta Y\right)\right].$$

Taking the derivative with respect to θ,

$$\frac{\partial}{\partial\theta} cE_{S_T}\left[\frac{f_{P_{T-1}}(Y)}{f_{S_{T-1}}(Y)}\left(\theta^2 - 2\theta Y\right)\right] = 0.$$

Thus,

$$c\left[E_{S_T}\left[\frac{f_{P_{T-1}}(Y)}{f_{S_{T-1}}(Y)}(2\theta)\right] - E_{S_T}\left[\frac{f_{P_{T-1}}(Y)}{f_{S_{T-1}}(Y)}2Y\right]\right] = 0.$$

So,

$$2\theta E_{S_T}\left[\frac{f_{P_{T-1}}(Y)}{f_{S_{T-1}}(Y)}\right] = 2E_{S_T}\left[\frac{f_{P_{T-1}}(Y)}{f_{S_{T-1}}(Y)}Y\right].$$

Therefore, the solution to the loss-minimization problem is

$$\theta = \frac{E_{S_T}\left[\frac{f_{P_{T-1}}(Y)}{f_{S_{T-1}}(Y)}Y\right]}{E_{S_T}\left[\frac{f_{P_{T-1}}(Y)}{f_{S_{T-1}}(Y)}\right]}.$$

5 Hajek Estimator

The general survey-sampling Hájek estimator is

$$\widehat{\bar{Y}}_H = \frac{\sum_{i\in s} Y_i/\pi_i}{\sum_{i\in s} 1/\pi_i},$$

where π_i is unit i's inclusion probability. It is a normalized inverse-probability-weighted mean.

In causal inference, the same estimator is applied separately to treated and control units:

$$\widehat{\mu}_1^H = \frac{\sum_i A_i Y_i/e_i}{\sum_i A_i/e_i}, \qquad \widehat{\mu}_0^H = \frac{\sum_i (1-A_i)Y_i/(1-e_i)}{\sum_i (1-A_i)/(1-e_i)},$$

and

$$\widehat{\tau}_H = \widehat{\mu}_1^H - \widehat{\mu}_0^H.$$

Here, A_i and Z_i both indicate treatment status. The notation Z_i is usually used for randomized treatment assignment, while A_i is often used for treatment or exposure in observational studies. In an RCT, the assignment probability is known from the randomization design. In an observational study, the propensity score

$$e_i = P(A_i = 1 \mid X_i, U_i)$$

must generally be estimated.

The paper *Design-Based Variance Estimation of the Hájek Effect Estimator in Stratified and Clustered Experiments*, by Xinhe Wang and Ben B. Hansen, studies the estimation of a weighted average treatment effect in stratified and cluster-randomized controlled trials. It estimates the treatment and control potential-outcome means using Hájek estimators and defines the treatment-effect estimator as their difference:

$$\widehat{\tau} = \widehat{\rho}_1 - \widehat{\rho}_0.$$

The authors show that this estimator can equivalently be obtained by minimizing an assignment-probability-weighted L_2 loss or by fitting a weighted least-squares regression. They then represent the same Hájek estimator as an M-estimator, which allows them to derive its large-sample behavior. In particular, they establish that $\widehat{\tau}$ is asymptotically normal around the true treatment effect τ. This result provides the basis for variance estimation, confidence intervals, and hypothesis testing.

For designs with many small strata, the authors show that direct-adjustment estimators, such as the Imai estimator and fixed-effects estimator, can be inconsistent when cluster sizes differ and treatment effects are associated with cluster size. This occurs because these methods first estimate effects within strata and can effectively give differently sized clusters equal weight, whereas the Hájek estimator directly targets the overall size-weighted treatment and control means.

The authors also propose a design-based variance estimator that works across strata of different sizes, including matched pairs with only one treated and one control cluster. This variance estimator supports the construction of standard errors, confidence intervals, and hypothesis tests. Their simulations show that the Hájek estimator is particularly advantageous in cluster-randomized trials with many small strata, unequal cluster sizes, and heterogeneous treatment effects that are correlated with cluster size.

6 Continuous to Binary Proof

For binary $Y \in \{0, 1\}$, the continuous-outcome estimator becomes

$$\begin{aligned} p_T &= \frac{E_{S_T}\left[\frac{Y}{\alpha_T(Y)}\right]}{E_{S_T}\left[\frac{1}{\alpha_T(Y)}\right]} \\ &= \frac{\frac{1\cdot\mathbb{P}_{S_T}(Y=1)}{\alpha_T(1)} + \frac{0\cdot\mathbb{P}_{S_T}(Y=0)}{\alpha_T(0)}}{\frac{\mathbb{P}_{S_T}(Y=1)}{\alpha_T(1)} + \frac{\mathbb{P}_{S_T}(Y=0)}{\alpha_T(0)}}. \end{aligned}$$

Let

$$s_T = \mathbb{P}_{S_T}(Y = 1).$$

Then

$$\mathbb{P}_{S_T}(Y = 0) = 1 - s_T.$$

Therefore,

$$p_T = \frac{\frac{s_T}{\alpha_T(1)}}{\frac{s_T}{\alpha_T(1)} + \frac{1-s_T}{\alpha_T(0)}}.$$

Multiplying the numerator and denominator by $\alpha_T(1)$, we get

$$p_T = \frac{s_T}{s_T + \frac{\alpha_T(1)}{\alpha_T(0)}(1 - s_T)}.$$

Define the binary response-bias ratio

$$\alpha = \frac{\alpha_T(1)}{\alpha_T(0)}.$$

Then

$$\begin{aligned} p_T &= \frac{s_T}{s_T + \alpha(1 - s_T)} \\ &= \frac{s_T}{\alpha + s_T - \alpha s_T}. \end{aligned}$$

7 Density Ratio Estimation

In our derivation above, it is clear that we will need a robust estimation of the density ratio between the survey distribution and the target population. It is generally suboptimal to estimate the ratio by first estimating the two component densities and then taking their ratio. Thus, we look to estimate the ratio directly.

We assume we are given i.i.d. samples

$$\{Y_i^{S_{T-k}}\}_{i=1}^{n_{S_{T-k}}}$$

from the survey distribution S_{T-k}, with density $f_{S_{T-k}}(y)$, and i.i.d. samples

$$\{Y_j^{P_{T-k}}\}_{j=1}^{n_{P_{T-k}}}$$

from the target population distribution P_{T-k}, with density $f_{P_{T-k}}(y)$. We aim to estimate

$$r(y) = \frac{f_{S_{T-k}}(y)}{f_{P_{T-k}}(y)}$$

from the samples

$$\{Y_i^{S_{T-k}}\}_{i=1}^{n_{S_{T-k}}} \qquad \text{and} \qquad \{Y_j^{P_{T-k}}\}_{j=1}^{n_{P_{T-k}}}.$$

In a setting in which we assume covariates, we have samples from the following distributions:

$$\begin{aligned} (Y_{j,T-k}^{P}) &\overset{i.i.d.}{\sim} P_{T-k}, \qquad j = 1, \ldots, n_{P_{T-k}}, \\ (Y_{i,T-k}^{S}) &\overset{i.i.d.}{\sim} S_{T-k}, \qquad i = 1, \ldots, n_{S_{T-k}}. \end{aligned}$$

A classical approach is to use a probabilistic classifier to estimate the density ratio. More specifically, we define a binary label that indicates whether a data point came from P_{T-k} or S_{T-k}:

$$A = \begin{cases} 1, & Y_i \text{ comes from } P_{T-k}, \\ 0, & Y_i \text{ comes from } S_{T-k}. \end{cases}$$

Then the density ratio can be written as

$$\begin{aligned} r(y) &= \frac{f_{S_{T-k}}(y)}{f_{P_{T-k}}(y)} \\ &= \frac{\mathbb{P}(y \mid A = 0)}{\mathbb{P}(y \mid A = 1)}. \end{aligned}$$

By Bayes' rule,

$$\mathbb{P}(y \mid A = 0) = \frac{\mathbb{P}(A = 0 \mid y)\mathbb{P}(y)}{\mathbb{P}(A = 0)},$$

and

$$\mathbb{P}(y \mid A = 1) = \frac{\mathbb{P}(A = 1 \mid y)\mathbb{P}(y)}{\mathbb{P}(A = 1)}.$$

Therefore,

$$\begin{aligned} r(y) &= \frac{\mathbb{P}(A = 0 \mid y)\mathbb{P}(y)/\mathbb{P}(A = 0)}{\mathbb{P}(A = 1 \mid y)\mathbb{P}(y)/\mathbb{P}(A = 1)} \\ &= \frac{\mathbb{P}(A = 1)\mathbb{P}(A = 0 \mid y)}{\mathbb{P}(A = 0)\mathbb{P}(A = 1 \mid y)}. \end{aligned}$$

Since

$$\mathbb{P}(A = 1 \mid y) = 1 - \mathbb{P}(A = 0 \mid y),$$

we have

$$r(y) = \frac{\mathbb{P}(A = 1)\mathbb{P}(A = 0 \mid y)}{\mathbb{P}(A = 0)\left(1 - \mathbb{P}(A = 0 \mid y)\right)}.$$

The class probabilities are

$$\mathbb{P}(A = 0) = \frac{n_{S_{T-k}}}{n_{P_{T-k}} + n_{S_{T-k}}},$$

and

$$\mathbb{P}(A = 1) = \frac{n_{P_{T-k}}}{n_{P_{T-k}} + n_{S_{T-k}}}.$$

$$\frac{1}{r(y)} = \frac{\mathbb{P}(A = 0)\left(1 - \mathbb{P}(A = 0 \mid y)\right)}{\mathbb{P}(A = 1)\mathbb{P}(A = 0 \mid y)}.$$

Plugging this into the self-normalized estimator gives

$$\hat{\mu}_T = \frac{\sum_{i=1}^{n_{S_T}} \frac{\mathbb{P}(A=0)\left(1-\widehat{\mathbb{P}}(A=0|Y_{i,T}^S)\right)}{\mathbb{P}(A=1)\widehat{\mathbb{P}}(A=0|Y_{i,T}^S)} Y_{i,T}^S}{\sum_{i=1}^{n_{S_T}} \frac{\mathbb{P}(A=0)\left(1-\widehat{\mathbb{P}}(A=0|Y_{i,T}^S)\right)}{\mathbb{P}(A=1)\widehat{\mathbb{P}}(A=0|Y_{i,T}^S)}}.$$

The factor

$$\frac{\mathbb{P}(A=0)}{\mathbb{P}(A=1)}$$

does not depend on i, so it cancels from the numerator and denominator. Therefore, the final estimator is

$$\hat{\mu}_T = \frac{\sum_{i=1}^{n_{S_T}} \frac{1-\widehat{\mathbb{P}}(A=0|Y_{i,T}^S)}{\widehat{\mathbb{P}}(A=0|Y_{i,T}^S)} Y_{i,T}^S}{\sum_{i=1}^{n_{S_T}} \frac{1-\widehat{\mathbb{P}}(A=0|Y_{i,T}^S)}{\widehat{\mathbb{P}}(A=0|Y_{i,T}^S)}}.$$

8 Statistical testing

Using the response-bias formulation,

$$\log(\alpha_t(x,y)) - \log(\alpha_t(x,y')) = \theta_t^T \left\{\eta(x,y') - \eta(x,y)\right\},$$

we can test whether the sampling-bias parameter is stable over time. The null model assumes that the same parameter θ applies across all time points:

$$H_0 : \theta_1 = \theta_2 = \cdots = \theta_m = \theta.$$

Thus, under the null model,

$$\log(\alpha_t(x,y)) - \log(\alpha_t(x,y')) = \theta^T \left\{\eta(x,y') - \eta(x,y)\right\}, \qquad t = 1, \ldots, m.$$

The full model allows the sampling-bias parameter to vary over time:

$$H_A : \text{at least one } \theta_t \text{ differs.}$$

Thus, under the full model,

$$\log(\alpha_t(x,y)) - \log(\alpha_t(x,y')) = \theta_t^T \left\{\eta(x,y') - \eta(x,y)\right\}, \qquad t = 1, \ldots, m.$$

The null model is nested inside the full model because it is the special case of the full model where

$$\theta_1 = \theta_2 = \cdots = \theta_m.$$

9 Importance Reweighting

I want to show in the binary case, we are unable to use importance reweighting to obtain the target quantity at time T. Importance reweighting says that:

$$E_{p_t}[y] = E_{s_t}[r_t(y) \cdot y]$$

$$r_t(y) = \frac{dp_t(y)}{ds_t(y)} = \frac{\imath_{p_t}(Y=1)}{\imath_{p_t}(Y=1|R=1)} = \frac{s_t}{\alpha + s_t - \alpha s_t} * \frac{1}{s_t} = \frac{1}{\alpha + s_t - \alpha s_t}$$

. In our setup, importance reweiting cannot be used to estimate the target quantity at time T using past data:

$$E_{p_t}[y] \neq E_{s_t}[r_{t-1}(y) \cdot y]$$

$$\begin{aligned} E_{s_t}[r_{t-1}(y) \cdot y] &= r_{t-1}(y) * 1 * \imath_{S_t}(Y=1) + r_{t-1}(y) * 0 * \imath_{s_t}(Y=0) \\ &= r_{t-1}(y)\imath_{s_t}(Y=1) = \frac{s_t}{\alpha + s_{t-1} - \alpha s_{t-1}} \end{aligned}$$

$$\frac{s_t}{\alpha + s_{t-1} - \alpha s_{t-1}} \neq \frac{s_t}{\alpha + s_t - \alpha s_t}$$

because $s_t \neq s_{t-1}$ thus, $E_{p_t}[y] \neq E_{s_t}[r_{t-1}(y) \cdot y]$

10 Synthetic Simulation

In the simulation, I assume the setting:

$$P_0 = Y \sim N(0,1)$$
$$P_1 = T \sim N(0.5,1)$$

$$\alpha(y) = \frac{1}{1+e^{-y}}$$

From these assumptions, we can get the closed form solutions for the density ratio and show that we obtain the true population mean at time t.

$$f_{P_0}(y) = \frac{f_{S_0}(y)\mathbb{P}_{P_0}(R=1)}{\alpha_0(y)}$$

$$\frac{f_{P_0}(y)}{f_{S_0}(y)} = \frac{f_{S_0}(y)\mathbb{P}_{P_0}(R=1)}{\alpha_0(y)} * \frac{1}{f_{S_0}(y)} = \frac{\mathbb{P}_{P_0}(R=1)}{\alpha_0(y)}$$

$\mathbb{P}_{P_0}(R=1)$ can be redefined as follows:

$$\begin{aligned} \mathbb{P}_{P_0}(R=1) &= \int_{-\infty}^{\infty} \alpha(y) \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(y-\theta)^2}{2\sigma^2}}\, dy \\ &= \int_{-\infty}^{0} \alpha(y) \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(y-0)^2}{2\sigma^2}}\, dy + \int_{0}^{\infty} \alpha(y) \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(y-0)^2}{2\sigma^2}}\, dy \\ &= \int_{\infty}^{0} \alpha(-y) \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(-y)^2}{2\sigma^2}}\, (-dy) + \int_{0}^{\infty} \alpha(y) \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(y)^2}{2\sigma^2}}\, dy \\ &= \int_{0}^{\infty} \alpha(-y) \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(-y)^2}{2\sigma^2}}\, dy + \int_{0}^{\infty} \alpha(y) \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(y)^2}{2\sigma^2}}\, dy \end{aligned}$$

$(-y)^2 = y^2$, thus we can make the following simplication:

$$\begin{aligned}
\int_0^\infty \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(y)^2}{2\sigma^2}}\ (\alpha(-y)+\alpha(y)) &= \int_0^\infty \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(y)^2}{2\sigma^2}}\ (\frac{1}{1+e^y}+\frac{1}{1+e^{-y}}) \\
&= \int_0^\infty \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(y)^2}{2\sigma^2}}\ (\frac{1}{1+e^y}+\frac{e^y}{e^y+e^y*e^{-y}}) \\
&= \int_0^\infty \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(y)^2}{2\sigma^2}}\ (\frac{1+e^y}{1+e^y}) \\
&= \int_0^\infty \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(y)^2}{2\sigma^2}} \\
&= \frac{1}{2}\int_{-\infty}^\infty \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(y)^2}{2\sigma^2}} \\
&= \frac{1}{2}\int_{-\infty}^\infty \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(y)^2}{2\sigma^2}} \\
&= \frac{1}{2}*1 \\
&= \frac{1}{2}
\end{aligned}$$

$$\frac{f_{P_0}(y)}{f_{S_0}(y)} = \frac{\mathbb{P}_{P_0}(R=1)}{\alpha_0(y)} = \frac{1}{2\alpha_0(y)}$$

Plugging this into the loss minimization, we can show that this closed form density ratio does indeed give us the true target quantity:

$$\theta = \frac{E_{S_T}\left[\frac{f_{P_{T-1}}(Y)}{f_{S_{T-1}}(Y)}Y\right]}{E_{S_T}\left[\frac{f_{P_{T-1}}(Y)}{f_{S_{T-1}}(Y)}\right]}$$

This loss minimizer then becomes

$$\theta^* = \frac{\mathbb{E}_{S_T}\left[\frac{1}{2\alpha_0(y)}Y\right]}{\mathbb{E}_{S_T}\left[\frac{1}{2\alpha_0(y)}\right]}.$$

Writing the expectations as integrals,

$$\theta^* = \frac{\int_{-\infty}^\infty y\frac{1}{2\alpha_0(y)} f_{S_T}(y)\,dy}{\int_{-\infty}^\infty \frac{1}{2\alpha_0(y)} f_{S_T}(y)\,dy}.$$

Since

$$f_{S_T}(y) = \frac{\alpha(y) f_{P_T}(y)}{Z_T},$$

the numerator is

$$\mathbb{E}_{S_T}[w_0(Y)Y] = \int_{-\infty}^\infty y\frac{1}{2\alpha_0(y)}\frac{\alpha(y) f_{P_T}(y)}{Z_T}\,dy.$$

$$\begin{aligned}\mathbb{E}_{S_T}[w_0(Y)Y] &= \frac{1}{2Z_T}\int_{-\infty}^{\infty} y f_{P_T}(y)\,dy \\ &= \frac{1}{2Z_T}\mathbb{E}_{P_T}[Y].\end{aligned}$$

Since

$$P_T = N(0.5, \sigma^2), \qquad \mathbb{E}_{P_T}[Y] = 0.5,$$

it follows that

$$\mathbb{E}_{S_T}[w_0(Y)Y] = \frac{0.5}{2Z_T}.$$

Similarly for the denominator,

$$\begin{aligned}\mathbb{E}_{S_T}[w_0(Y)] &= \int_{-\infty}^{\infty} \frac{1}{2\alpha_0(y)} \frac{\alpha(y) f_{P_T}(y)}{Z_T}\,dy \\ &= \frac{1}{2Z_T}\int_{-\infty}^{\infty} f_{P_T}(y)\,dy \\ &= \frac{1}{2Z_T}.\end{aligned}$$

Thus,

$$\begin{aligned}\theta^* &= \frac{\displaystyle\int_{-\infty}^{\infty} y \frac{1}{2\alpha_0(y)} f_{S_T}(y)\,dy}{\displaystyle\int_{-\infty}^{\infty} \frac{1}{2\alpha_0(y)} f_{S_T}(y)\,dy} \quad = \frac{\frac{0.5}{2Z_T}}{\frac{1}{2Z_T}} \\ &= 0.5.\end{aligned}$$

11 Related Work

In the covariate-free setup, if we assume that the simple response-bias parameter α_t does vary with time, then we need to forecast the bias parameter at the current time T, α_T, in order to calculate the target quantity p_T utilizing s_T.

Giannone, Reichlin, and Small (2008), "Nowcasting: The Real-Time Informational Content of Macroeconomic Data," explores a nowcasting method to nowcast current-quarter GDP when official GDP is observed with delay, but many monthly indicators are released asynchronously. They introduce a dynamic factor model that handles real-time datasets and updates the nowcast as new information arrives. When observations are missing, the paper uses a Kalman smoother to estimate latent factors.

Another approach is to use state-space models, which are useful when the quantity of interest evolves over time but is not directly observed without noise. Durbin and Koopman "Time Series Analysis by State Space Methods" presents a reference for state-space modeling in which observations are decomposed into latent components such as trends, regression effects, seasonal components, and disturbances. The Kalman filter then provides a recursive way to estimate these latent states as new observations arrive. In this setting, α_t is not directly observed at every time point, but for past periods where s_t and p_t are observed, we can estimate it and treat α_t as a latent time-varying state. We can then forecast it using past noisy estimates.